

**Time:** 5 minutes**Outcome:** Identify Minimum Efficient Scale & Constant Returns to Scale.**Difficulty:**

Intermediate

Minimum Efficient Scale & Constant Returns to Scale



THE CORE IDEA

Minimum efficient scale is the smallest output at which long-run average cost reaches its minimum. Before MES, additional scale can still lower long-run average cost; at MES the minimum has been reached. MES summarizes how much output a firm must reach before it captures all available scale economies.



HOW TO RECOGNIZE IT

- Find the lowest point of the long-run average-cost curve.
- Minimum efficient scale is the smallest output at which that minimum cost is attained.
- A flat minimum indicates a range of outputs with constant returns to scale.
- Do not confuse a flat average-cost segment with output remaining unchanged.

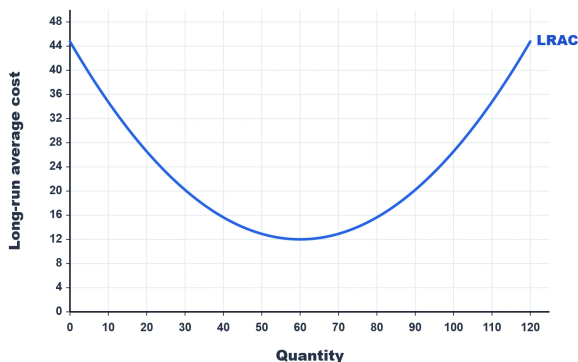


WATCH OUT

A common mistake is to invent flat LRAC region. The graph has a single rounded minimum rather than a visibly flat minimum-cost interval.



WORKED EXAMPLE: FINDING MINIMUM EFFICIENT SCALE



The LRAC curve reaches its lowest value, about \$12 per unit, at an output of 60. Minimum efficient scale is the smallest output that attains minimum long-run average cost, so the graph's MES is 60 units. Outputs below 60 have not exhausted economies of scale.



CHECK YOURSELF

If LRAC first reaches its minimum at 5,000 units and remains flat until 8,000, MES is:?

**READY?**

Return to the game and master this concept.